

Problem 1

The moment generating function (MGF) of an exponential random variable ξ with parameter λ is

$$\begin{aligned} M_{\xi}(t) &= \mathbb{E}[e^{t\xi}] \\ &= \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx \\ &= \frac{\lambda}{\lambda - t} - \frac{\lambda e^{(t-\lambda)x}}{\lambda - t} \Big|_0^{\infty} \\ &= \frac{\lambda}{\lambda - t} \end{aligned}$$

for any $t < \lambda$.

So the MGF of S is

$$\begin{aligned} M_S(t) &= \mathbb{E}[e^{tS}] \\ &= \mathbb{E}\left[e^{t \sum_{i=1}^n \xi_i}\right] \\ &= \prod_{i=1}^n \mathbb{E}[e^{t\xi_i}] \\ &= \prod_{i=1}^n M_{\xi_i}(t) \\ &= \left(\frac{\lambda}{\lambda - t}\right)^n \end{aligned}$$

for $t < \lambda$, which is the MGF of distribution $\Gamma(n, \lambda)$. Thus the distribution of S is $\Gamma(n, \lambda)$.

Problem 2

$$\begin{aligned} \mathbb{E}[V(X|Y)] &= \mathbb{E}_Y[\mathbb{E}_X[X^2|Y] - (\mathbb{E}_X[X|Y])^2] \\ &= \mathbb{E}_X[X^2] - \mathbb{E}_Y[(\mathbb{E}_X[X|Y])^2], \\ V(\mathbb{E}[X|Y]) &= \mathbb{E}_Y[(\mathbb{E}[X|Y])^2] - (\mathbb{E}_X[X])^2 \\ &= \mathbb{E}_Y[(\mathbb{E}[X|Y])^2] - (\mathbb{E}_X[X])^2. \end{aligned}$$

Thus,

$$\text{RHS} = \mathbb{E}[V(X|Y)] + V(\mathbb{E}[X|Y]) = \mathbb{E}_X[X^2] - (\mathbb{E}_X[X])^2 = V(X) = \text{LHS}.$$

Problem 3

$$\begin{aligned}
\mathbb{E}[L] &= \mathbb{E}[N + 8] = \mathbb{E}[N] + 8 = 10, \\
\text{Var}(L) &= \text{Var}(N + 8) = \text{Var}(N) = 1.6, \\
\mathbb{E}[L^2] &= (\mathbb{E}[L])^2 + \text{Var}(L) = 101.6, \\
\mathbb{E}[T] &= \mathbb{E}_L[\mathbb{E}_T[T|L]] \\
&= \mathbb{E}_L[2L - 4] \\
&= 2\mathbb{E}[L] - 4 = 16, \\
\text{Var}(T) &= \mathbb{E}[\text{Var}(T|L)] + \text{Var}(\mathbb{E}[T|L]) \\
&= \mathbb{E}\left[\frac{L(L-1)}{4}\right] + \text{Var}(2L - 4) \\
&= \frac{1}{4}(\mathbb{E}[L^2] - \mathbb{E}[L]) + 4\text{Var}(L) = 29.3.
\end{aligned}$$

Problem 4

$$\mathbb{E}(X|Y) = aY + b \Rightarrow \mathbb{E}X = a\mathbb{E}Y + b \quad \text{and} \quad \mathbb{E}(XY|Y) = aY^2 + bY$$

$$\begin{aligned}
\text{Var}(X + Y) &= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y) \\
&= \mathbb{E}X^2 - (\mathbb{E}X)^2 + \mathbb{E}Y^2 - (\mathbb{E}Y)^2 + 2\mathbb{E}XY - 2(\mathbb{E}X)(\mathbb{E}Y)
\end{aligned}$$

$$\begin{cases} 5 = 7a + b \\ 36.6 = 51.4a + 7b \end{cases} \Rightarrow \begin{cases} a = \frac{2}{3} \\ b = \frac{1}{3} \end{cases}$$

Problem 5

(a)

$$\begin{aligned}
\mathbb{P}(\text{Bob wins}) &= \sum_{n=1}^{\infty} \mathbb{P}(\text{Bob wins in } n\text{th game}) \\
&= \sum_{n=1}^{\infty} \mathbb{P}(\text{1st all the same}, \dots, \text{n-1th all the same, nth Bob win}) \\
&= \sum_{n=1}^{\infty} \left(\frac{1}{4}p + \frac{1}{4}(1-p)\right)^{n-1} \times \left(\frac{1}{4}p + \frac{1}{4}(1-p)\right) \\
&= \sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^n \\
&= \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{1}{3}
\end{aligned}$$

(b)

Let X be the number of rounds that they will toss

$$\begin{aligned}\mathbb{P}(X = n) &= \mathbb{P}(\text{1st all the same}, \dots, \text{n-1th all the same, nth sb. win}) \\ &= \left(\frac{1}{4}\right)^{n-1} \times \left(\frac{1}{4}p + \frac{1}{4}(1-p) + 2 \times \left(\frac{1}{2} \times \frac{1}{2}p + \frac{1}{2} \times \frac{1}{2}(1-p)\right)\right) \\ &= \left(\frac{1}{4}\right)^n \times 3\end{aligned}$$

Thus

$$\begin{aligned}\mathbb{E}X &= \sum_{n=1}^{\infty} n \times 3 \times \left(\frac{1}{4}\right)^n \\ &= 3 \times \sum_{n=1}^{\infty} n \times \left(\frac{1}{4}\right)^n \\ &= 3 \times \frac{\frac{1}{4}}{(1 - \frac{1}{4})^2} \\ &= \frac{3}{4} \times \frac{1}{(\frac{3}{4})^2} \\ &= \frac{4}{3}\end{aligned}$$

Problem 6

Assume p be the unknown probability of tossing a Head, Y be the outcome of the procedure.

$$\begin{aligned}P(Y = \text{Head}) &= P(\text{the first coin is Head} | \text{the outcomes differ}) \\ &= \frac{P(\text{the first coin is Head, the outcomes differ})}{P(\text{the outcomes differ})} \\ &= \frac{P(\text{the first coin is Head, the second coin is Tail})}{2P(H)P(T)} \\ &= \frac{p(1-p)}{2p(1-p)} \\ &= \frac{1}{2}\end{aligned}$$